

A two-stage personalized recommendation based on multi-objective teaching–learning-based optimization with decomposition

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ABSTRACT

Due to its successful application in information filtering and knowledge retrieval systems in the era of big data, personalized recommender system plays a significant role in meeting personalized demands of people from big data and has become a hot research hotspot. Traditional recommender systems only guarantee the recommendation accuracy and the recommendation diversity will be lost. In this paper, we propose a two-stage personalized recommendation (TSPR) algorithm based on an improved collaborative filtering (ICF) and multi-objective teaching–learning-based optimization (MOTLBO/D) with decomposition. Firstly, ICF made use of each user's preference and social neighborhood information to obtain a candidate recommendation list for each target user. In the proposed ICF method, the jump relationship between users and their unrated items was mined to describe the characteristic behavior information of users more fully. Moreover, MOTLBO/D was adopted to simultaneously optimize accuracy and diversity of recommendation list of items for each target user. In the proposed MOTLBO/D algorithm, the learner representation strategy is designed based on attributes of the recommendation problem, and the learners are updated by an improved TLBO procedure. Finally, the simulation results on two MovieLens datasets show that TSPR is effective and efficient in personalized recommender systems.

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1. Introduction

With the development of Internet, the volume of data become more and more large, and human beings enter the era of big data. How to effectively realize information filtering and knowledge retrieval from big data becomes a huge challenge for people. To deal with this information overload problem in various domains such as movies, news, and e-commerce and so on, personalized recommender systems (PRS) become important and indispensable tools as useful and effective data filtering and retrieval tools in various domains and have been getting more and more attention from researchers. PRS seek to make appropriate personalized recommendations for users by automatically learning users' personalized preferences [1]. Recent years, researchers have paid their attentions to PRS and developed various kinds of successful algorithms. These PRS can be classified three major categories: content-based (CB) methods, collaborative filtering (CF) methods and hybrid-based methods. CB methods try to discover the correlation between items by analyzing their attributes or contents based on

items' metadata information and then recommend similar items to users based on their previous choices [2–6]. Collaborative filtering (CF) methods try to predict users' preferences by analyzing the similar measurements of users and items based on the rating relationship between users and items and then recommend similar items to users based on their previous ratings [7–11]. Hybrid-based methods merge multiple recommendation techniques [12–16]. CF is one of the most fundamental and well-known personalized recommender methods owing to its simplicity and more advantages compared with the other two recommendation techniques.

Personalized recommender systems recommend items with higher ratings to users, and how to obtain appropriate recommendations is always a challenging task. Researchers have introduced them into personalized recommendation systems on the basis of the successful application of machine learning [17–20] and intelligence optimization [21–25] in many real-world fields. On the one hand, personalized recommender systems can be considered as a reduced problem of the prediction task by predicting ratings offline for all unrated items by the user, and the latent factor model such as Matrix Factorization (MF) [26] and Singular Value Decomposition (SVD) [27] were used to predict ratings. On the other hand,

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it can be considered as an optimization problem by maximizing the accuracy as much as possible. Hence, to improve the performance of PRS, many evolutionary algorithms (EAs) and swarm intelligence algorithms (SIAs) have been utilized to optimize the recommended lists of items by maximizing the accuracy. Rad and Lucas utilized an invasive weed optimization algorithm find the optimal priorities for each user [28]. Al-Shamri and Bharadwaj proposed a hybrid fuzzy-genetic method to PRS. In the method, the hybrid features of users and items were used to build a hybrid user model, a novel fuzzy distance function was developed to match different users with fuzzy models and genetic algorithm (GA) was utilized to obtain optimal features' weights [29]. Ju and Xu presented a new CF-based recommendation approach [30]. In the method, k-means clustering method was used to cluster users and artificial bee colony algorithm was used to search the optimal value of center points. Bakshi et al utilized K-means to improve scalability and accuracy of recommender systems and particle swarm optimization (PSO) to balance the weight of prediction ratings from the global and local neighbors [31]. Katarya and Verma employed a CF-based method with PSO [32]. In the method, users in the dataset was partitioned according to features of movies and the initial centers are obtained based on K-means clustering, and then the final clusters are obtained by FCM clustering algorithm with PSO. In addition, they also developed personalized recommender systems with other population-based optimization algorithms [33–36]. Gohari et al presented a semantic-enhanced trust-based PRS [37]. In the method, the optimal trust paths in the trust network are obtained by using ant colony optimization (ACO) and then the best neighbors of each user were selected to obtain the better recommendation performance. Choudhary et al developed a recommender system with gravitational search algorithm [38]. Yadav et al presented an improved CF-based recommender system with bat algorithm [39]. In the method, the bat algorithm was used to optimize the weights of items so as to find better recommendations. Katarya proposed a hybrid personalized recommender system with k-means clustering [40]. In the method, artificial bee colony was utilized to optimize the results of the k-means procedure. Logesh et al presented a CF-based PRS with quantum-behaved PSO (QPSO) [41]. In the method, QPSO was utilized to cluster the users in the dataset. Parvin et al developed a trust-aware CF-based personalized recommender system in which ACO is utilized to optimize the weights of users similar to the target user [42]. For more details on various recommender systems based on EAs and SIAs, one may be able to find in Refs. [43,44].

Although many EAs and SIAs have been successfully applied in personalized recommender systems, it may not be sufficient to consider recommendation accuracy alone [45]. To obtain appropriate personalized recommendations for users in personalized recommender systems, in addition to the accuracy metric, some other metrics such as the diversity, the novelty and so on should also be considered [46,47]. However, accuracy focuses on recommending a list of appropriate items to users that match users' past preference, whereas diversity focuses on recommending a list of items to users that are different from the users' past choices. They are conflicting performance metrics. To obtain a good balance of the accuracy and the diversity/novelty of recommendation lists, many multi-objective optimization (MOO) methods have been introduced into personalized recommender systems. Tyagi and Bharadwaj employed multi-objective PSO to extract only interesting association rules in CF-based recommendation system [48]. Zuo et al adopted NSGA-II to obtain optimal accuracy and diversity recommendation lists of items for users [49]. Ribeiro et al developed a new Pareto efficiency concept into MOEA to handle PRS to so as obtain optimal recommendation lists of items for users [50]. Geng et al developed a non-dominated neighbor

immune-algorithm-based recommender system to obtain a unique recommendation list with optimal accuracy and diversity to each target user [51]. Wang et al developed a CF-based MOEA for long tail recommendation to obtain optimal accurate and novel recommendation lists of items for users [52]. Ding et al employed a multi-objective cloud service recommendation algorithm to simultaneously consider accuracy and diversity and improve user satisfaction and formulate [53]. Karabadi et al employed a MOEA-based recommendation system to maximize both similarities with the active user and diversities between its members [54]. Lacerda employed a multi-objective ranked bandit was presented by presented to obtain good accurate, diverse and novel recommendation lists of items [55]. Ouni et al adopted NSGA-II to deal with software library recommendation so as to find a trade-off between three objectives [56]. Cui et al employed a novel probabilistic MOEA for personalized recommendation problems to obtain a good balance of accuracy and diversity of recommendation lists [57]. Chai et al. developed a new recommender system based on singular value decomposition (SVD) and multi-objective immune algorithm (MOIA) [58]. In the method, SVD was used to obtain a candidate recommendation list for each target user and then MOIA is used to obtain a candidate recommendation list for each target user. Cao et al a novel a two-layer distributed parallel cooperative coevolutionary MOEA to simultaneously optimize two conflicting performance metrics in terms of accuracy and diversity [59]. Mao et al developed multi-objective method based on the hypergraph ranking model for e-commerce recommendations [60]. Zapotecas-Martínez et al adopted a decomposition-based multi-objective method with grey wolf optimizer to obtain optimal recommendation lists of items for users [61]. Geng et al presented a MOIA for personalized location recommendation so as to optimize the matching qualities of similarity and geographic properties [62]. Xu developed a new improved CF-based multi-objective PRS for big-data oriented recommendation based on MapReduce framework [63].

As a population-based optimization algorithm, teaching-learning-based optimization (TLBO) has characters of simple concept and without the specific parameters, and it has been successfully applied many areas [64]. In this work, we mainly focus on MOO method for personalized recommender systems. That is, we choose to use decomposition-based MOO method with teaching-learning-based optimization (MOTLBO/D) to obtain a unique recommendation list for each target user. Our goal is to balance accuracy and diversity of recommender system and provide a recommendation method with higher accuracy and more diversity. We adopted a new two-stage recommendation strategy to generate top-N recommendation lists for users. In the rough recommender stage, an improved CF (ICF) recommendation method by taking account into preference and social neighborhood information was introduced to obtain a candidate recommendation list for each target user. In fact, in a PRS, when neighbors of a target user recommend a specific item, this target user may also be interested in this same item. In the refined recommender stage, a decomposition-based MOEA method was adopted to simultaneously optimize two conflicting performance metrics in terms of accuracy and diversity. Our proposed MOEA is based on TLBO. In MOTLBO/D, the learner representation strategy is designed based on attributes of the recommendation problem, and the learners are updated by an improved TLBO procedure.

Correspondingly, the main contributions of this work are as follows:

- (1) We proposed an improved CF-based method (ICF) with a novel similarity computational process by taking account into preference and social neighborhood information.

- (2) We proposed discrete TLBO framework is designed based on attributes of the recommendation problem and extend it to a new decomposition-based multi-objective variant.
- (3) We can obtain a set of recommendation lists with different accuracies and diversities to meet different demands of the decision maker through a single running.
- (4) Our experimental results indicate that our proposed TSPR method can obtain a good balance of multiple performance metrics on a classical MovieLens data and is a feasible and promising method.

The remainder of this paper is organized as follows: Section 2 describes the related works. In Section 3, a two-stage personalized recommendation (TSPR) algorithm based on ICF and MOTLBO/D is proposed. Section 4 provides experimental results of TSPR on two data sets. Finally, conclusions are presented in Section 5.

2. Background knowledge

In this section, we briefly discuss personalized recommendation problem, objective functions for evaluating recommended results and backtracking search algorithm.

2.1. Personalized recommendation problem

Personalized recommender systems (PRS) seek to predict users' preferences and make appropriate personalized recommendation lists of items for users. Assume that the score matrix *Rating* is the ratings of a set of items by a set of users. Considering the sparse problem of the original data in personalized recommender systems, there are few ratings in *Rating*. That is, many users did not give ratings for many items. Hence, personalized recommendation problems (PRPs) can be considered as: predict the unknown ratings of each item for each user and select some items with higher rating to recommend to the corresponding user.

To obtain a more popular or relevant list of items, items with higher ratings should be recommended to users. To obtain a more diverse or novel list of items, items with lower the intra-user diversity should be recommended to users. In our proposed TSPR algorithm, two conflicting maximization objective functions (i.e., the accuracy and diversity) are simultaneously optimized so as to obtain good recommendation performance in terms of accuracy and diversity. These two functions are described in detail as follows.

The accuracy function is utilized to evaluate the accuracy of recommendation items and it can guide the optimization process of the recommendation algorithm to obtain a set of appropriate items that can better satisfy users. Just like other methods [51,58], we use a matching function to define the accuracy of the recommendation list. In the paper, for the u -th user, it can be expressed as follow

$$F_1 = \frac{1}{\|R(u)\|} \sum_{i \in R(u)} \left(\frac{1}{\|P(u)\|} \sum_{j \in P(u)} \text{sim}(i, j) \right) \quad (1)$$

where $R(u)$ and $\|R(u)\|$ is the recommendation list of the u -th user and its length, $\|R(u)\|$ is the length of this list $R(u)$, $P(u)$ is a list of rated items by the u -th user, $\|P(u)\|$ is the length of this list $P(u)$, $\text{sim}(i, j)$ is the similarity of the i -th item from $R(u)$ and the j -th item from $P(u)$ according to the cosine-based similarity [65]. A higher value corresponds to a better accuracy.

The diversity function is utilized to evaluate the diversity of recommendation items and it can guide the searching process of the recommendation algorithm to contain more different recommendation items for each user. In the paper, it is represented by the

sum of item non-similarity of recommendation items. For the u -th user, it can be expressed as follow [66]

$$F_2 = 1 - \frac{\sum_{i \in R(u), j \in R(u), i \neq j} \text{sim}(i, j)}{\|R(u)\|(\|R(u)\| - 1)} \quad (2)$$

where $R(u)$ is the recommendation list of the u -th user, $\|R(u)\|$ is the length of this list $R(u)$, $\text{sim}(i, j)$ is the similarity of the i -th item and the j -th item from $R(u)$. A higher value corresponds to a better diversity.

2.2. Decomposition-based multi-objective optimization

Zhang and Li developed a novel decomposition-based MOEA framework (MOEA/D) in 2007 [67]. MOEA/D utilized a decomposition strategy based on traditional aggregation approaches to divide a multi-objective problem into a specific number of single-objective problems, and then these single optimization problems can be simultaneously optimized by using evolutionary algorithm and swarm intelligence optimization algorithms. Here several ways can be utilized to execute this division. Generally speaking, the most popular one is the Tchebycheff decomposition and this approach is used to solve the single scalar optimization problem shown as follows [67]:

$$\min g(x|w, z^*) = \max_{i \in \{1, \dots, m\}} \{w_i |f_i(x) - z_i^*|\}$$

where m is the number of problems to be optimized in objective variable space, $w = (w_1, \dots, w_m)$ is a weight vector subject to $w_i \geq 0$ and $\sum_{i=1}^m w_i = 1$. $z^* = (z_1^*, \dots, z_m^*)$ refers to the reference point in objective variable space subject to $z_i^* = \min \{f_i(x) | x \in \Omega\}$.

3. The proposed TSPR algorithm

In this section, we first give the framework the proposed two-stage personalized recommendation (TSPR) algorithm. Second, we introduce a new improved CF (ICF) recommendation method based on the jump relationship between users and their unrated items to obtain a candidate recommendation list for each target user. Finally, will give a detailed description about decomposition-based MOO method with teaching-learning-based optimization (MOTLBO/D) for PRPs.

3.1. Framework of the proposed algorithm

Recently, CF-based multi-objective algorithms have been successfully utilized to deal with PRPs. In this paper, we develop a two-stage personalized recommendation algorithm (TSPR) based on ICF recommendation method and MOTLBO/D to deal with the PR problem. The procedure of TSPR includes two main phases: rough recommender phase and refined recommender phase. The rough recommender stage utilizes the ICF recommendation method by taking account into preference and social neighborhood information to obtain a candidate recommendation list of items for each target user, and the refined recommender stage, MOTLBO/D was adopted to simultaneously optimize two conflicting performance metrics in terms of accuracy and diversity. The flowchart of the proposed TSPR algorithm shows in Fig. 1.

3.2. Rough recommender phase

CF-based methods try to predict users' preferences by analyzing the similar measurements of users and items according to the rating behaviors of users to items and then recommend similar items to users according to their previous ratings. However, the rating behaviors of users, such as whether the user scores or likes the

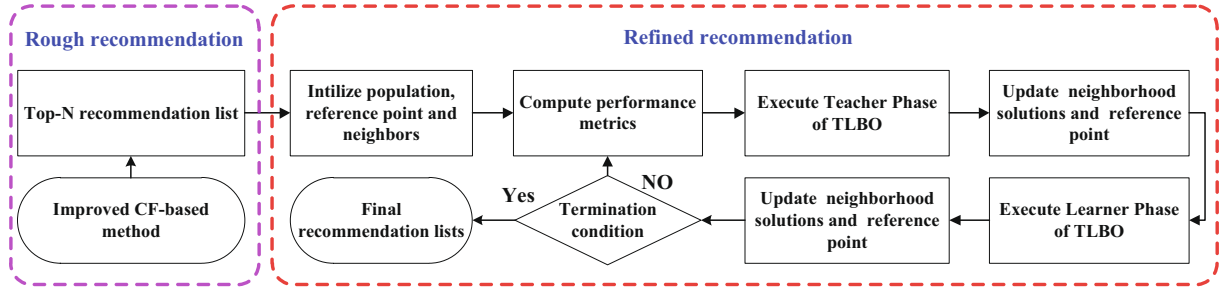


Fig. 1. Flowchart of the proposed TSPR algorithm.

item or so on, cannot fully reflect all behavior information of users. In addition to the direct rating behaviors of users to items, there are also different degrees of correlation between non-directly connected users and items. For example, a user can reach a certain unrated item through the limited jump of the user's common rated items so that a user has a certain correlation with each unrated item. Hence, the direct rating behaviors of users to the items can only simply present direct rating behavior relationships to rated items, and cannot express the relationship between users and their unrated items, and hence affecting the accuracy of recommendation list of items for each target user. In order to mine the jump relationship between users and their unrated items and describe the characteristic behavior information of users more fully, an improved CF-based method is developed to obtain a top-ranked candidate recommendation list in the rough recommender phase (Fig. 2).

The jump relationship can derive from different users' ratings on common rated items. Assume there are four users and five items for a recommender system. The rating matrix of this system is given as follow:

$$R = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \quad (4)$$

Through this rating matrix, we can construct the information network of user-to-item connection:

It can be observed that, User A rated Item 1 and Item 3. User A can reach Item 4 and Item 5 through 3 hops ($A \rightarrow 1 \rightarrow D \rightarrow 4$ and $A \rightarrow 3 \rightarrow C \rightarrow 5$), and it also can reach Item 2 through 5 hops ($A \rightarrow 1 \rightarrow D \rightarrow 4 \rightarrow B \rightarrow 2$). Similarly, User B rated Item 2 and Item 4. User B can reach Item 1 through 3 hops ($B \rightarrow 4 \rightarrow D \rightarrow 1$), Item 3 through 5 hops ($B \rightarrow 4 \rightarrow D \rightarrow 1 \rightarrow A \rightarrow 3$), and Item 5 through 7 hops ($B \rightarrow 4 \rightarrow D \rightarrow 1 \rightarrow A \rightarrow 3 \rightarrow C \rightarrow 5$). User C rated Item 3 and Item 5. User C can reach Item 1 through 3 hops ($C \rightarrow 3 \rightarrow A \rightarrow 1$), Item 2 through 7 hops ($C \rightarrow 3 \rightarrow A \rightarrow 1 \rightarrow D \rightarrow 4 \rightarrow B \rightarrow 2$), and Item 4 through 5 hops ($B \rightarrow 3 \rightarrow A \rightarrow 1 \rightarrow D \rightarrow 4$). User D rated Item 1 and Item 4. User C can reach Item 2 and Item 3 through 3 hops ($D \rightarrow 4 \rightarrow B \rightarrow 2$ and $D \rightarrow 1 \rightarrow A \rightarrow 3$), and Item 5 through 5 hops ($D \rightarrow 1 \rightarrow A \rightarrow 3 \rightarrow C \rightarrow 5$). From these jumps, it can be seen that the number of jumps reflects the similarity among users to a certain extent.

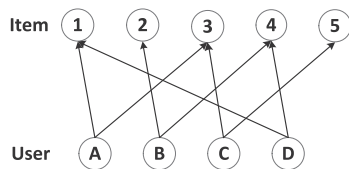


Fig. 2. Information network of user-to-item connection.

Need to note that, if User u has multiple paths to Item i , we choose the path with the least number of jumps. Based on the analysis of the jumping times between users and their unrated items, the behavior characteristic matrix of users based on jumping times is established as follows.

$$C_{ui} = \begin{cases} 1 & \text{User } u \text{ rated Item } i \\ e^{-\alpha(k-1)} & \text{User } u \text{ unrated Item } i \end{cases} \quad (5)$$

where k is the jumping times that User u can reach Item i , α is the attenuation coefficient and it is a random within the range (0,0.5).

In traditional CF-based methods, the unknown ratings of each item by each user are predicted according to the similarity between each user or each item and the known ratings of each item by each user. In our ICF-based method, taking account into the jump relationship between users and their unrated items, the predicted rating of User u to Item i can be calculated as follows

$$P_{ui} = \frac{\sum_{k \in R, k \neq i} S_{ik} \cdot C_{uk} \cdot R_{uk}}{\sum_{k \in R, k \neq i} S_{ik} \cdot C_{uk}} \quad (6)$$

where P_{ui} is the predicted rating according to traditional CF-based methods, S_{ik} is the similarity between Item i and Item k , C_{uk} the behavior characteristic component of User u to Item k , and R is the candidate recommendation item set for the target user u .

Then, N items with higher predicted ratings are considered as a candidate recommendation list for the target user u .

3.3. Refined recommender phase

In the refined recommender phase, a decomposition-based MOEA method with TLBO (MOTLBO/D) was adopted to simultaneously optimize accuracy and diversity of recommendation list for each target user. To deal with personalized recommendation problems, TLBO needs to be extended to a discrete TLBO (DTLBO). The description of the extended DBSA is given in detail as follows.

3.3.1. Individual representation and initialization

For the PR problem, the goal is to obtain a recommendation list of items with the optimal performance. Hence, for solving the PR problem by using MOTLBO/D, each learner of the population represents a recommendation list of items. It is natural and intuitive to represents a recommendation list of items by using a set of real integer values in which each real integer value corresponds to a unique item. That is, the i -th learner can be expressed as a real integer vector $X_i = (x_{i1}, x_{i2}, \dots, x_{ij}, \dots, x_{iN})$, and here where N is the number of the candidate recommendation items for a specific target user. In our method, each element x_{ij} is encoded as 0 or 1. That is, if the j -th item is recommended to the target user, x_{ij} is encoded as 1; otherwise, x_{ij} is encoded as 0. It needs to be noted that, the learner should be satisfied with the following constraints

$$K = \sum_{i=1}^N x_{ij} \quad (7)$$

where K is the length of the final recommendation list of items, and N is the length of the candidate recommendation list of items.

Assume that there is a candidate recommendation list with fifteen items, and the length of the final recommendation list of items is set to 6, then the learner can be represented shown as Fig. 3.

3.3.2. Teacher and mean individual update rule

In order to deal with the PR problem, the best learner (*Teacher*) and the existing mean individual (*Mean*) needs to be obtained before executing teacher phase of TLBO. Different from the continuous version of TLBO, here *Teacher* and *Mean* are all integer vectors in which each element corresponds to a unique item. Obviously, if the average value of each dimension of all learners is directly rounded to an integer so as to obtain the mean individual *Mean* of all learners, this mean individual *Mean* may not satisfy the constraints of Eq. (7). That is, the representation of the mean individual *Mean* may be illegal in this case. Hence, the mean individual *Mean* should be revised. The executing process is as follows. When the average value of each dimension of all learners is obtained, the average values of all dimensions are sorted according to descending order. Then, the K (i.e., the number of the final recommendation items for a target user) gene positions with the largest values are selected, and the values in these corresponding gene positions are set to 1. An example illustration of the mean individual *Mean* is shown in Fig. 4.

In addition, it needs to be noted that, in our proposed MOTLBO/D, for each learner, his or her *Teacher* and *Mean* are updated within his or her neighborhood.

3.3.3. Teacher phase of DTLBO

In the *Teacher Phase*, each learner makes use of the difference between *Teacher* and *Mean* of all learners to learn from them and generate a new trial learner. However, each learner is expressed as a real integer vector, and the updating strategies of the continuous version of TLBO no longer fit this discrete representation. Therefore, the updating strategies of TLBO must be redefined to match personalized recommendation problems. As previous explained, the implementation steps of the *Teacher Phase* in DTLBO are given in the following.

Algorithm 1: Teaching ()

Step 1: Search through the best learner *Teacher_i* and the mean individual *Mean_i* in the neighborhood of the learner X_i to find gene positions where the elements of these gene positions are simultaneously equivalent to 1, and then retain these gene positions and set the rest to 0 so as to generate a temporary learner *A* in our DTLBO variant.

Step 2: Search through the learner X_i to find gene positions where the elements are equivalent to 1 and at the same time they are not included in *A*, and then retain these gene positions and set the rest to 0 so as to generate a temporary vector *B*.

Step 3: Search through *Teacher_i* and *Mean_i* to find gene positions where the elements are not simultaneously equivalent to 1 and at the same time they are not included in *A* and *B*, and then retain these gene positions and set the rest to 0 so as to generate a temporary vector *C*.

Step 4: Execute a new individual generation operation. If the number of 1 s in the temporary vector *A* is equal to K (K is the length of the final recommendation list of items), the temporary

vector *A* is considered as the new learner *newX_i*. If the number of 1 s in the temporary vector *A* is L ($L < K$, K is the number of the final recommendation items for a target user), clone all L gene positions where the elements are equal to 1 to corresponding gene positions in the new learner *newX_i*, and then randomly select $K-L$ gene positions where the elements are equal to 1 from the temporary vector *B* or the temporary vector *C* to corresponding gene positions in the new learner *newX_i*.

Step 5: Duplicate solutions exist in the original TLBO algorithm, and it is necessary to modify them to maintain diversity in the population and avoid falling into the local optima. Hence, we introduce a random mutation operation into DTLBO. That is, randomly select two gene positions where the elements are not simultaneously equivalent to 1 or 0, and then exchange two elements to corresponding gene positions.

An example illustration of *Teacher Phase* is shown in Fig. 5.

3.3.4. Learner phase of DTLBO

In the *Learner Phase*, the learner X_i makes use of the difference between himself/himself and the other randomly selected learner X_k of all learners in the class to generate a new trial learner. As previous explained, the implementation steps of the *Learner Phase* in DTLBO are given in the following.

Algorithm 2: Learning()

Step 1: Search through X_i and X_k to find gene positions where the elements are simultaneously equivalent to 1, and then retain these gene positions and set the rest to 0 so as to generate a temporary learner *A* in our DTLBO variant.

Step 2: Search through the learner X_i to find gene positions where the elements are equivalent to 1 and at the same time they are not included in *A*, and then retain these gene positions and set the rest to 0 so as to generate a temporary vector *B*.

Step 3: Search through the learner X_k to find gene positions where the elements are not simultaneously equivalent to 1 and at the same time they are not included in *A* and *B*, and then retain these gene positions and set the rest to 0 so as to generate a temporary vector *C*.

Then, Steps 4–5 in *Teacher Phase* are executed to generate a new learner. An example illustration of *Teacher Phase* is shown in Fig. 6.

Based on the above DTLBO, we developed a new decomposition-based MOEA method (MOTLBO/D). In our MOTLBO/D, neighbors of each learner are obtained according to the distances among their corresponding aggregation their weight vectors of all objective, and then each learner and her/his neighbors form a subpopulation. Each subpopulation is optimized and updated according to the above DTLBO. Need to note that, in our TSPR framework, the discrete variants of other population-based optimization algorithms can be utilized in the refined phase.

4. Experimental studies

To verify the feasibility and effectiveness of the proposed TSPR approach, we execute experiments on the classical Movielens 100 K dataset and Movielens 1 M dataset. The recommendation performance metrics of TSPR in terms of precision and novelty are compared with those of representative Item-based CF (I-CF) [7], User-based CF (U-CF) [7] and matrix factorization method (MF) [26].

4.1. Data set and experimental settings

We assess the effectiveness of TSPR on the classical Movielens 100 K dataset and Movielens 1 M dataset (It can be downloaded from <https://grouplens.org/datasets/movielens/>). Movielens 100 K dataset consists of 943 users, 1682 movies and 100,000 ratings.

Dimension	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Learner	0	0	1	0	0	1	0	0	1	0	0	1	1	0	1

Fig. 3. An example of learner representation.

Dimension	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\frac{1}{N} \sum_{i=1}^N \text{Learner}_i$	0.32	0.04	0.27	0.02	0.41	0	0.12	0.11	0.24	0.01	0.08	0.60	0.30	0.08	0.20
Mean	1	0	1	0	1	0	0	0	1	0	0	1	1	0	0

Fig. 4. An example of the mean individual Mean representation.

Dimension	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
X_i	0	0	1	0	0	1	0	0	1	0	0	1	1	0	1
Teacher _i	1	0	0	1	0	0	1	1	0	0	1	0	0	0	1
Mean _i	1	0	0	0	0	1	1	0	1	0	0	0	1	0	1
A	1	0	0	0	0	0	1	0	0	0	0	0	0	0	1
B	0	0	1	0	0	1	0	0	1	0	0	1	1	0	0
C	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0
newX _i	1	0	1	0	0	0	1	1	0	0	1	0	0	0	1
	Mutation  Operation														
newX _i	1	0	1	0	0	0	1	0	0	0	1	0	1	0	1

Fig. 5. An example illustration of Teacher Phase.

Dimension	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
X_i	0	0	1	0	0	1	0	0	1	0	0	1	1	0	1
X_k	1	0	0	1	1	0	0	1	0	0	1	0	0	0	1
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
B	0	0	1	0	0	1	0	0	1	0	0	1	1	0	0
C	1	0	0	1	1	0	0	1	0	0	1	0	0	0	0
newX _i	1	0	0	1	0	1	0	1	0	0	0	1	0	0	1
	Mutation  Operation														
newX _i	1	0	1	0	0	0	1	0	0	0	1	0	0	1	1

Fig. 6. An example illustration of Learner Phase.

Movielens 1 M dataset consists of 6040 users, 3952 movies and 1,000,209 ratings. In our experiments, the training data set randomly extracts 80% from the whole data set and the testing data set extracts remaining 20% from the whole data set.

In our experiments, the simulations are carried out on MATLAB R2018b under a PC system with an Intel® Core™ i7-9700 CPU with 8 GB RAM. In our proposed TSPR, the population size was 100, the number of neighbors of each individual was 20, the number of iterations was 500, and the lengths of the candidate recommendation list and the final recommendation list were 50 and 10, respectively. For all recommender algorithms on all datasets, the number of the simulations are set to 20.

4.2. Evaluation metrics

Generally, different metrics are used to assess the recommendation performance of a PR algorithm. In this paper, we adopt widely

used metrics to measure accuracy and diversity [41,57] of our proposed TSPR algorithm.

Precision: Precision is a well-known evaluation metric to assess the accuracy performance of the final recommendation items and it can be expressed as follows.

$$P_u = \frac{|R_u \cap T_u|}{|R_u|} \quad (8)$$

where R_u is the final recommendation item set for the target user u in the testing data set, $|R_u|$ is the number of items in R_u , T_u is items which is rated 4 or higher by the target user u in the testing data set, $R_u \cap T_u$ is the set of items which belong to both R_u and T_u , and $|R_u \cap T_u|$ is the number of items in $R_u \cap T_u$. The recommendation list with the greater the value P_u has the better accuracy performance.

Diversity: Diversity is the most widely used evaluation metric to assess the differences among all items from recommendation list of items. The well-known intra-user diversity is often used to assess the diversity of the final recommendation items and it can be expressed as follows.

$$D_u = \frac{1}{|R_u|(|R_u| - 1)} \sum_{\substack{i,j \in R_u \\ i \neq j}} S(i,j) \quad (9)$$

where R_u is the final recommendation item set for the target user u in the testing data set, $|R_u|$ is the number of items in R_u , and $S(i,j)$ is the similarity between item i and item j of the item set R_u . The recommendation list with smaller the value D_u has the better diversity performance.

Novelty: Novelty is an evaluation metric to assess the unpopularity of items in the final recommendation list and it can be expressed as follows.

$$N_u = \frac{1}{|R_u|} \sum_{i \in R_u} d_i \quad (10)$$

where R_u is the final recommendation item set for the target user u in the testing data set, $|R_u|$ is the number of items in R_u , and d_i is the number of users that rated the item i of the item set R_u . The recommendation list with smaller the value N_u has the better novelty performance.

Need to note that, to assess the recommendation performances of algorithms for all users in Movielens 100 K dataset, statistical means P , D and N are used and they are the average values of P_u , D_u and N_u based on each user u over a run, respectively.

4.3. Experimental results

As we described above, our proposed TSPR can obtain a number of solutions (i.e., a set of recommendation lists of items) for each target user only through a single running.

Fig. 7 and Fig. 8 gives the obtained Pareto front (PF) of the first, the fifth and tenth users after TSPR has been run once. In Fig. 7 and Fig. 8, Accuracy value is evaluated according to Eq. (1) and Diversity value is evaluated according to Eq. (2). Every point in Fig. 7 and Fig. 8 refers to a set of recommendation list of items for the

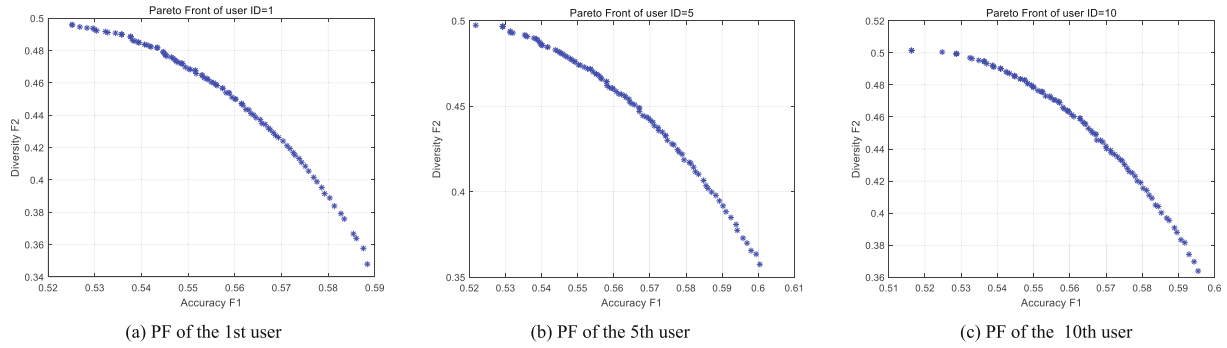


Fig. 7. Pareto front of different users obtained by TSPR on Movielens 100 K dataset.

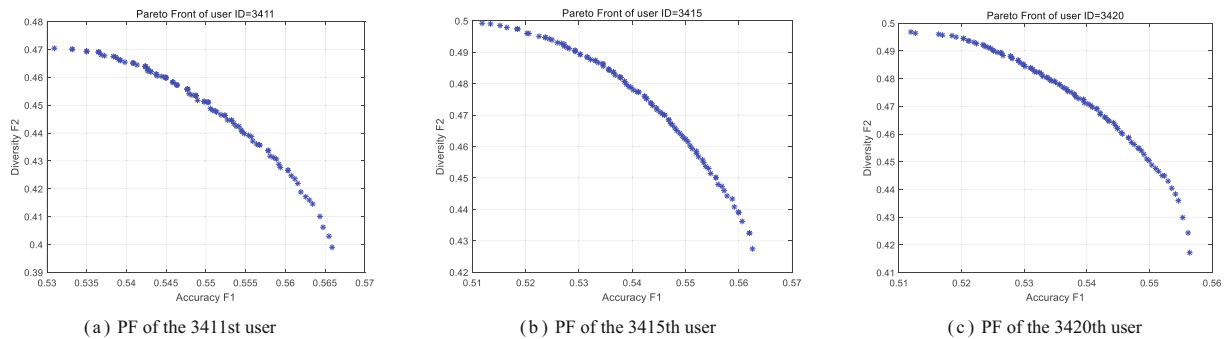


Fig. 8. Pareto front of different users obtained by TSPR on Movielens 1 M dataset.

specific target user, and points with higher accuracy values have smaller diversity values.

Tables 1–6 report the values of precision, diversity and novelty metrics of different algorithms on ten different target users, respectively. For Pareto front solution set, we can't evaluate the quality of any two solutions. Hence, to access our proposed TSPR, the maximum values and minimum values of the particular target user in terms of precision, diversity and novelty metrics are selected, and then their average values over 20 runs are respectively calculated for the particular target user. The terms 'Min', 'Max' and 'Mean' represent the minimum value, the maximum value and the mean value of the correspond performance metric for all recommendation lists, respectively. The best results among all algorithms are shown in boldface.

In terms of precision metric, Table 1 and Table 2 respectively show the precision values obtained by different algorithms of ten target users from two Movielens datasets. It can be seen from Table 1 that, for Movielens 100 K dataset, seven users' maximum values obtained by our algorithm TSPR are the best with the largest precision value. I-CF obtain the best performance for user 7, U-CF obtain the best performance for user 8 and MF obtain the best performance for user 2. It can also be seen from Table 2 that, for Movielens 1 M dataset, five users' maximum values obtained by our algorithm TSPR are the best with the largest precision value. Among these users, TSPR and I-CF have the same value for the user 3416, TSPR and U-CF have the same value for the user 3418 and TSPR and MF have the same value for the users 3412 and 3413. I-CF obtain the best performance for user 3414 and MF obtain the best performance for users 3411 and 3420. In addition, TSPR can predict their preferences which the other comparison methods cannot predict, such as users 3 and 9 in Movielens 100 K dataset and the user 3419 in Movielens 1 M dataset. It can be concluded that the lists given by our TSPR algorithm may not have the more

accurate, but it has the ability to give a recommendation that traditional methods cannot find.

In terms of diversity metric, Table 3 and Table 4 respectively show the diversity values obtained by different algorithms of ten target users from two Movielens datasets. For Movielens 100 K dataset, it can be seen from Table 3 that, the mean values obtained by our algorithm TSPR are the best with the smallest diversity value and are better than the values of the other comparison methods. Besides, all minimum values are smaller than those of other methods for all ten users. For Movielens 1 M dataset, it can be also observed from Table 4 that, the mean values obtained by our algorithm TSPR are better than those of the other comparison methods for users 3411 and 3415, and the minimum values are smaller than those of other methods for five users 3411, 3414, 3415 and 3418. MF is better than our algorithm for users 3412, 3413, 3414, 3417 and 3420, and I-CF is better than our algorithm for user 3419. It can be concluded that, although the lists given by our TSPR algorithm may not have the more accurate for all users, but it has the ability to give a good diversity recommendation that traditional methods cannot find.

Table 5 and Table 6 also respectively show the novelty values obtained by different algorithms of ten target users from two Movielens datasets. It can also be observed from Table 5 that, for Movielens 100 K dataset, the minimum values obtained by our algorithm TSPR are the best with the smallest novelty value for six users. The maximum values obtained by our algorithm TSPR are better than those obtained by I-CF and U-CF for users 7–9, and the mean values obtained by our algorithm are better than the values obtained by U-CF except for user 10. I-CF performs the best for users 2, 3 and 6, and MF performs the best for user 4. For Movielens 1 M dataset, it can also be observed from Table 6 that the minimum values obtained by our algorithm TSPR are the best with the smallest novelty value for four users 3411, 3415,

Table 1

Precision metric of different algorithms on ten target users from Movielens 100 K dataset.

User ID		1	2	3	4	5	6	7	8	9	10
I-CF		0.200	0	0	0.04	0.040	0.180	0.500	0	0	0.280
U-CF		0.140	0.020	0	0.04	0.020	0.240	0.260	0.060	0	0.100
MF		0.100	0.060	0	0.02	0.020	0.100	0.220	0	0	0.060
TSPR	Min	0	0	0	0	0	0	0.020	0	0	0.000
	Max	0.320	0	0.020	0.040	0.040	0.300	0.270	0.020	0.040	0.320
	Mean	0.166	0	0	0.020	0.018	0.180	0.056	0.002	0.012	0.116

Table 2

Precision metric of different algorithms on ten target users from Movielens 1 M dataset.

User ID		3411	3412	3413	3414	3415	3416	3417	3418	3419	3420
I-CF		0	0	0	0.300	0	0.200	0.100	0	0	0
U-CF		0	0.100	0	0	0	0.100	0	0.200	0	0.100
MF		0.600	0.200	0.300	0.100	0	0.100	0.100	0.100	0	0.300
TSPR	Min	0	0	0	0	0	0	0	0	0	0.100
	Max	0.280	0.200	0.300	0.200	0	0.200	0	0.200	0.100	0.200
	Mean	0.120	0.060	0.120	0.080	0	0.080	0	0.080	0.020	0.120

Table 3

Diversity metric of different algorithms on ten target users from Movielens 100 K dataset.

User ID		1	2	3	4	5	6	7	8	9	10
I-CF		0.4478	0.4487	0.4527	0.4546	0.4722	0.4856	0.4849	0.5019	0.5079	0.4973
U-CF		0.4626	0.4788	0.4623	0.4653	0.4698	0.4695	0.4609	0.4786	0.4762	0.4628
MF		0.4405	0.4358	0.4429	0.4352	0.4327	0.4453	0.4367	0.4581	0.4410	0.4338
TSPR	Min	0.4251	0.4142	0.4200	0.4189	0.4255	0.4170	0.4209	0.4197	0.4240	0.4144
	Max	0.4396	0.4493	0.4381	0.4345	0.4503	0.4371	0.4445	0.4441	0.4488	0.4458
	Mean	0.4313	0.4305	0.4294	0.4270	0.4355	0.4272	0.4328	0.4324	0.4359	0.4274

Table 4

Diversity metric of different algorithms on ten target users from Movielens 1 M dataset.

User ID		3411	3412	3413	3414	3415	3416	3417	3418	3419	3420
I-CF		0.4900	0.4537	0.4247	0.4865	0.4307	0.4808	0.4367	0.4634	0.4221	0.4481
U-CF		0.4376	0.4334	0.4223	0.4248	0.4368	0.4248	0.4317	0.4272	0.4334	0.4298
MF		0.4618	0.4854	0.4647	0.4906	0.4729	0.4809	0.4758	0.4767	0.4966	0.4774
TSPR	Min	0.4250	0.4456	0.4267	0.4293	0.4250	0.4216	0.4335	0.4235	0.4226	0.4303
	Max	0.4327	0.4599	0.4392	0.4395	0.4379	0.4442	0.4459	0.4310	0.4398	0.4627
	Mean	0.4281	0.4544	0.4315	0.4346	0.4311	0.4308	0.4387	0.4274	0.4327	0.4440

3416 and 3418. In addition, the minimum values obtained by our algorithm are better than the values obtained by I-CF for five users 3411, 3414, 3415, 3416 and 3418, and the values obtained by U-CF for four users 3411, 3413, 3415, 3416, 3418, 3419 and 3420, the values obtained by MF for all ten users. I-CF performs the best for four users 3413, 3417, 3419 and 3420 and U-CF performs the best for two users 3412 and 3414. From the above discussions, it can be seen that our proposed TSPR algorithm doesn't perform well in most cases, the reason for this problem is that MOTLBO/D optimizer need to design a suitable search process to improve the optimization performance.

Fig. 9 and Fig. 10 report the performance curves of the precision, diversity and novelty metrics of different algorithms on all target users from two Movielens datasets. For Pareto front solution set, we can't evaluate the quality of any two solutions. Hence, to access our proposed TSPR, the largest values of statistical means for all target users in terms of accuracy function and diversity function are selected, and then their mean values over 20 runs are respectively calculated and marked with as TSPR-A and TSPR-D. TSPR-M represents the mean of all the recommendation lists for all target users.

It can be seen from Fig. 9 that for Movielens 100 K dataset, in term of the precision metric of recommendation result, TSPR-A

has the best performance with the largest precision value, and TSPR-D has the worst performance with the smallest precision value. In term of the diversity metric of recommendation result, TSPR-D has the best performance with the smallest diversity value, MF is better than I-CF and U-CF obtains the worst performance with the largest diversity value. In term of the novelty metric of recommendation result, MF has the best performance with the smallest novelty value, TSPR is worse than MF but better than other two algorithms. It can also be seen from Fig. 10 that for Movielens 1 M dataset, in term of the precision metric of recommendation result, TSPR-A has the best performance with the largest precision value, and TSPR-D has the worst performance with the smallest precision value. In term of the diversity metric of recommendation result, TSPR-D has the best performance with the smallest diversity value, U-CF is better than MF, and I-CF obtains the worst performance with the largest diversity value. In term of the novelty metric of recommendation result, TSPR-A has the best performance with the smallest novelty value and TSPR-D has the worst performance. On the whole, TSPR improves precision, diversity and novelty the of recommendation lists to meet the various needs of users and has better optimization effect on novelty, and hence offering more options for decision makers.

Table 5

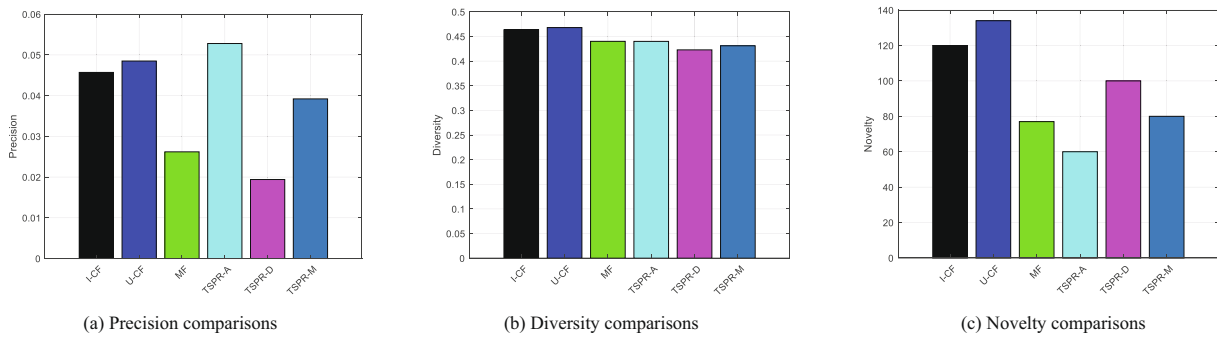
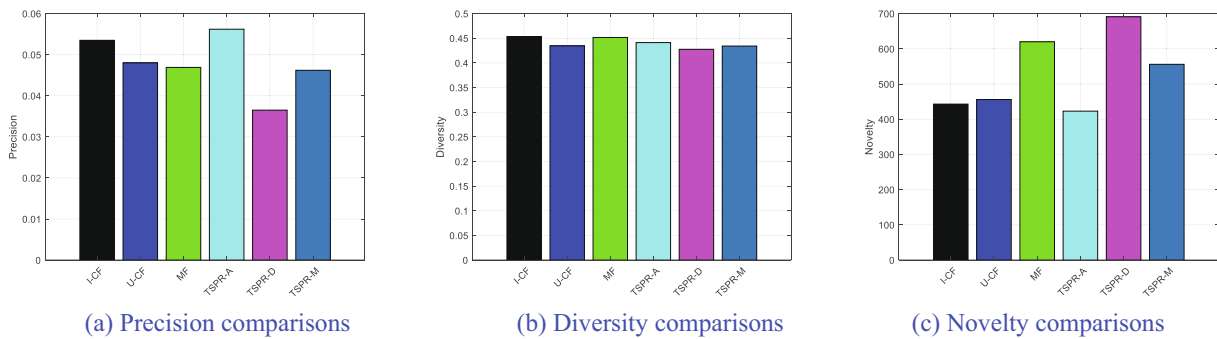
Novelty metric of different algorithms on ten target users from Movielens 100 K dataset.

User ID		1	2	3	4	5	6	7	8	9	10
I-CF		87	54	32	61	77	42	127	134	185	96
U-CF		91	124	156	139	138	132	112	134	135	113
MF		73	78	91	56	79	56	71	103	76	41
TSPR	Min	58	57	53	60	59	56	42	58	47	41
	Max	105	101	119	98	128	109	111	102	108	116
	Mean	85	80	82	81	87	81	79	81	81	75

Table 6

Novelty metric of different algorithms on ten target users from Movielens 1 M dataset.

User ID		3411	3412	3413	3414	3415	3416	3417	3418	3419	3420
I-CF		467	364	251	689	254	1127	106	409	172	172
U-CF		375	262	481	163	435	339	406	427	636	526
MF		849	773	734	885	775	861	898	883	736	775
TSPR	Min	289	587	427	470	192	273	541	180	332	256
	Max	655	1085	725	693	491	1014	768	501	654	899
	Mean	464	801	582	587	412	629	617	355	563	602

**Fig. 9.** Performance comparisons of different algorithms on Movielens 100 K dataset.**Fig. 10.** Performance comparisons of different algorithms on Movielens 1 M dataset.

4.4. Comparison with NNIA-RS

In this subsection, we further compare the proposed TSPPR with NNIA-RS on Movielens 1 M dataset. The values of precision, diversity and novelty metrics of different algorithms on ten different target users are shown in Table 7, respectively. The best results among all algorithms are shown in boldface.

In terms of precision metric, it can also be observed from Table 7 that, the maximum values obtained by our TSPPR algorithm are better than the values obtained by NNIA-RS for four users 3412, 3413, 3418 and 3420. The mean precision of NNIA-RS is significantly better for five users 3411, 3412, 3414, 3415 and 3417, and the mean precision of TSPPR is better than those of NNIA-RS for five users 3413, 3416, 3418, 3419 and 3420. In terms of diversity metric,

the minimum values obtained by TSPPR are better for four users 3412, 3413, 3415 and 3420. The maximum diversity of TSPPR is significantly better for all users, and the mean diversity of TSPPR is significantly better for five users 3411, 3412, 3413, 3415 and 3420. The minimum values obtained by NNIA-RS are better for five users 3411, 3414, 3416, 3417 and 3418. In terms of novelty metric, compared with NNIA-RS, the minimum novelty of TSPPR is better for three users, the maximum novelty of TSPPR is better for five users, and the mean novelty of TSPPR is better for four users. From the above discussions, it can be seen that it can obtain some good results that NNIA-RS cannot obtain, although not all metric values obtained by our proposed TSPPR algorithm are better than those of NNIA-RS in some cases. And at the same time, it means that our MOTLBO/D optimizer need to design a suitable search process to

Table 7
Metrics on Movielens 1 M dataset.

User ID	Precision						Diversity						Novelty					
	TSPR			NNIA-RS			TSPR			NNIA-RS			TSPR			NNIA-RS		
	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
3411	0	0.280	0.120	0.1	0.4	0.200	0.425	0.433	0.428	0.381	0.486	0.438	289	655	464	637	944	765
3412	0	0.200	0.060	0	0.1	0.093	0.446	0.460	0.454	0.448	0.488	0.466	587	1085	801	226	751	482
3413	0	0.300	0.120	0	0.1	0.093	0.427	0.439	0.432	0.464	0.488	0.473	427	725	582	350	903	536
3414	0	0.200	0.080	0.1	0.5	0.255	0.429	0.440	0.435	0.311	0.487	0.402	470	693	587	351	1059	645
3415	0	0	0	0.1	0.2	0.135	0.425	0.438	0.431	0.476	0.487	0.482	192	491	412	646	1304	948
3416	0	0.200	0.080	0	0.2	0.035	0.422	0.444	0.431	0.271	0.482	0.371	273	1014	629	44	929	344
3417	0	0	0	0.1	0.3	0.165	0.434	0.446	0.439	0.358	0.489	0.435	541	768	617	97	627	352
3418	0	0.200	0.080	0	0	0	0.424	0.431	0.427	0.159	0.488	0.329	180	501	355	348	736	522
3419	0	0.100	0.020	0	0.1	0.018	0.423	0.440	0.433	0.237	0.481	0.365	332	654	563	90	359	181
3420	0.100	0.200	0.120	0	0	0	0.430	0.463	0.444	0.448	0.487	0.466	256	899	602	159	439	315

improve the optimization performance compared with the NNIA algorithm for personalized recommendation problems.

5. Conclusion

We focus on how to improve the recommendation performance of personalized recommender system by making use of a two-stage personalized recommendation (TSPR) algorithm in this paper. We introduce an improved CF recommendation method to obtain a candidate recommendation list for each target user, and then adopt MOTLBO/D to simultaneously optimize accuracy and diversity of recommendation list of items for each target user. Finally, the experimental simulations on two Movielens datasets show that TSPR is effective and efficient in improving the recommendation performance, especially in terms of diversity. However, there is still much work for improvement of novelty. In addition, to control the dimension of the decision variables, i.e., the length a candidate recommendation list of items, here only a small number of items have been extracted by using an improved CF method as the candidate recommendation list of items. Moreover, TSPR only obtains a set of recommendation list of items for one target user through a single running, and it need spend more running times for all target users. Hence, we will concentrate on how to more accurately select the candidate recommendation list of items. How to use faster recommendation techniques and parallel MOEA optimizers with more better optimization performances will also be our further work.

CRedit authorship contribution statement

Feng Zou: Conceptualization, Methodology, Visualization, Writing - original draft. **Debao Chen:** Investigation, Funding acquisition. **Qingzheng Xu:** Validation. **Ziqi Jiang:** Software. **Jiahui Kang:** Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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